Started on	Wednesday, 11 March 2026, 10:05 PM
State	Finished
Completed on	Wednesday, 11 March 2026, 10:05 PM
Time taken	5 secs
Grade	**0.00** out of 100.00

Question 1

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:

```
area = radius * radius * π
volume = area * length
```

Hint: Assume pi = 3.14159.

For example:

Input	Result
5.5 12	Enter the radius and length of a cylinder: 5.5 12.0 The area is 95.0330975 The volume of the cylinder is 1140.39717
22.12 4.65	Enter the radius and length of a cylinder: 22.12 4.65 The area is 1537.162394096 The volume of the cylinder is 7147.805132546401

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise02_02 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    // Enter radius of the cylinder
    System.out.print("Enter the radius and length of a cylinder: ");
    double radius = input.nextDouble();
    double length = input.nextDouble();
    System.out.print(radius + " ");
    System.out.println(length);

    // Compute area and volume
    double area = radius * radius * 3.14159;
    double volume = area * length;

    // Display result
    System.out.println("The area is " + area);
    System.out.println("The volume of the cylinder is " + volume);
  }
}
```

```
23
```

Question 2

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that reads the subtotal and the gratuity (tip) rate, then computes the gratuity and total.

For example, if the user enters **10** for subtotal and **15%** for gratuity rate, the program displays **$1.5** as gratuity and **$11.5** as total.

For example:

Input	Result
10 15	Enter subtotal and gratuity rate: 10.0 15.0 The gratuity is $1.5 total is $11.5

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise02_05 {
  public static void main(String[] args) {
    // Read subtotal
    java.util.Scanner input = new java.util.Scanner(System.in);
    System.out.print("Enter subtotal and gratuity rate: ");
    double subtotal = input.nextDouble();
    double rate = input.nextDouble();

    System.out.println(subtotal + " " + rate);

    double gratuity = subtotal * rate / 100;
    double total = subtotal + gratuity;

    System.out.println("The gratuity is $" + gratuity +
        " total is $" + total);
  }
}
```

Question 3

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that prompts the user to enter the minutes (e.g., 1 billion), and displays the number of years and days for the minutes.

Hint: Assume a year has **365** days.

For example:

Input	Result
1000000000	Enter the number of minutes: 1000000000 1000000000 minutes is approximately 1902 years and 214 days

Answer: (penalty regime: 0 %)

```
1
```

▼ **Show/hide question author's solution (Java)**

```java
import java.util.Scanner;

public class Exercise02_07 {
  public static void main(String[] args) {
    // Prompt the user to enter the number of minutes
    Scanner input = new Scanner(System.in);
    System.out.print("Enter the number of minutes: ");
    long minutes = input.nextLong();
    System.out.println(minutes);

    long numberOfDays = minutes / (24 * 60);
    long numberOfYears = numberOfDays / 365;
    long remaingDays = numberOfDays % 365;

    // Display results
    System.out.println(minutes + " minutes is approximately " +
      numberOfYears + " years and " + remaingDays + " days");
  }
}
```

Question 4

Not answered

Mark 0.00 out of 15.00

Write a JAVA program that calculates the energy needed to heat water from an initial temperature to a final temperature.

Your program should prompt the user to enter the amount of water in kilograms and the initial and final temperatures of the water.

The formula to compute the energy is

Q = **M** * (finalTemperature – initialTemperature) * 4184

where **M** is the weight of water in kilograms, temperatures are in degrees **Celsius**, and energy **Q** is measured in joules.

For example:

Input	Result
55.5 3.5 10.5	Enter the amount of water in kilograms: 55.5 Enter the initial temperature: 3.5 Enter the final temperature: 10.5 The energy needed is 1625484.0

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
public class Exercise02_10 {
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);

    System.out.print("Enter the amount of water in kilograms: ");
    double mass = input.nextDouble();
    System.out.println(mass);

    System.out.print("Enter the initial temperature: ");
    double initialTemperature = input.nextDouble();
    System.out.println(initialTemperature);

    System.out.print("Enter the final temperature: ");
    double finalTemperature = input.nextDouble();
    System.out.println(finalTemperature);

    double energy = mass * (finalTemperature - initialTemperature) * 4184;

    System.out.print("The energy needed is " + energy);
  }
}

```

Question 5

Not answered

Mark 0.00 out of 20.00

Write a JAVA program that reads an integer between **0** and **1000** and adds all the digits in the integer.

For example, if an integer is 932, the sum of all its digits is 14.

Hint: Use the **%** operator to extract digits, and use the **/** operator to remove the extracted digit. For instance, 932 % 10 = 2 and 932 / 10 = 93.

For example:

Input	Result
999	`Enter an integer between 0 and 1000: 999` `The sum of all digits in 999 is 27`

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
// Exercise02_06.java: Summarize all digits in an integer < 1000
public class Exercise02_06 {
  // Main method
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);
    // Read a number
    System.out.print("Enter an integer between 0 and 1000: ");
    int number = input.nextInt();
    System.out.println(number);

    // Find all digits in number
    int lastDigit = number % 10;
    int remainingNumber = number / 10;
    int secondLastDigit = remainingNumber % 10;
    remainingNumber = remainingNumber / 10;
    int thirdLastDigit = remainingNumber % 10;

    // Obtain the sum of all digits
    int sum = lastDigit + secondLastDigit + thirdLastDigit;

    // Display results
    System.out.println("The sum of all digits in " + number + " is " + sum);
  }
}

```

Question 6

Not answered

Mark 0.00 out of 20.00

Write a JAVA program that reads in investment amount, annual interest rate, and number of years, and displays the future investment value using the following formula:

$$\text{futureInvestmentValue} =$$

$$\text{investmentAmount} \times (1 + \text{monthlyInterestRate})^{\text{numberOfYears}*12}$$

For example, if you enter amount **1000**, annual interest rate **3.25%**, and number of years **1**, the future investment value is **1032.98**.

For example:

Input	Result
1000.56 4.25 1	Enter the investment amount: 1000.56 Enter annual interest rate: 4.25 Enter number of years as an integer: 1 Future value is $1043.92

Answer: (penalty regime: 0 %)

```
1
```

Show/hide question author's solution (Java)

```java
// Exercise02_11.java: Create a method for computing future value
public class Exercise02_21 {
  public static void main(String[] args) {
    java.util.Scanner input = new java.util.Scanner(System.in);

    // Enter the investment amount
    System.out.print("Enter the investment amount: ");
    double investmentAmount = input.nextDouble();
    System.out.println(investmentAmount);

    // Enter yearly interest rate
    System.out.print("Enter annual interest rate: ");
    double annualInterestRate = input.nextDouble();
    System.out.println(annualInterestRate);

    // Obtain monthly interest rate
    double monthlyInterestRate = annualInterestRate / 1200;

    // Enter number of years
    System.out.print("Enter number of years as an integer: ");
    int numOfYears = input.nextInt();
    System.out.println(numOfYears);

    double futureValue = investmentAmount * Math.pow(1 + monthlyInterestRate, numOfYears * 12);
```

```
26      System.out.print("Future value is $" + (int)(futureValue * 100) / 100.0);
27    }
28  }
29
```